

The Modernization of Monitorization: Technological Adaption and Innovation as Mechanisms for College Athlete Surveillance

Journal of Sport and Social Issues
2025, Vol. 49(2–3) 175–195
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DOI: 10.1177/01937235251330206
journals.sagepub.com/home/jss



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Abstract

Athletes competing in National Collegiate Athletic Association (NCAA) athletics have long been monitored by their respective athletic departments. As technology continues to advance and become more socially pervasive, many NCAA member institutions have issued wearable technology devices to athletes. The present study sought to examine the function of such implementation of wearable technology in collegiate athletics. Findings from interviews with current NCAA athletes reveal the widespread use of wearable technology in collegiate athletics has resulted in the increased surveillance of athletes. Such surveillance was most notably mentioned with regards to athletic performance tracking, invasive sleep tracking, and general location tracking. In addition, participants stated that while implementation of wearable technology within training programs was instrumental in improving performance, such increasing biometric monitoring was detrimental to individual development and occasionally fostered dissension among teammates. Accordingly, this research calls into question the use and function of wearable technology in collegiate athletics.

Keywords

NCAA, total institution, wearable technology, hegemony, paternalism

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Technological innovation occurs daily and is wholly pervasive within global society. Perpetually in pursuit of a competitive edge, the sport industry has welcomed technological innovations for their competitive, performance, and economic benefits (Bourdon et al., 2017; Cardinale & Varley, 2017; Luczak et al., 2020; Ringuet-Riot et al., 2013). In fact, many wearable technologies are designed and marketed around their use for sport specific activities (James & Petrone, 2016; Ringuet-Riot et al., 2013). Among National Collegiate Athletic Association (NCAA) member institutions wearable technology has become a commonly integrated component of athletic-specific programming. Accordingly, college athletes are being provided and are utilizing wearable technology at rates never seen before (Dodd, 2014; Luczak et al., 2020).

A common wearable technology implemented by athletic departments across the NCAA is *WHOOP*. As of the beginning of the 2023–2024 academic year, WHOOP has existent partnerships with numerous NCAA member institutions, spanning Divisions I, II, and III (Gettysburg College, 2020; Holmes, 2020; Lemire, 2022). A performance sport band similar to a watch, WHOOP provides the ability to track users' heart rate, location, sleep patterns, and other biometric markers. Indicative of the perceived value of wearable technology, Pennsylvania State University (Penn State) head football coach James Franklin has attributed WHOOP to improvements in athletic training programming within the program. Specifically focused on sleep monitoring, Franklin posts "players' sleep results daily, by position group, so they can see their data and adjust their schedules accordingly" (Snyder, 2018, para. 9). Other NCAA coaches have explicitly noted the impact wearable technologies have had within athletic programs (Dodd, 2014). Given the relationship between technological innovation and athletic training (Bourdon et al., 2017; Cardinale & Varley, 2017; Dodd, 2014; Luczak et al., 2020), coaches and staff rely on wearable technologies such as WHOOP for the stated purpose of maximizing an athlete's performance.

The use of wearable technology among college athletes, however, has led to some legal scrutiny (Arnold & Sade, 2017; Brown & Brown, 2021; Jessop & Baker, 2019; Seshadri et al., 2019; Smolenski, 2019; Studnicka, 2020). Wearable technologies ostensibly distributed to NCAA athletes for performance improvement have been utilized at times to surveil college athletes' non-performance activity. Geo-location monitoring, sleep tracking, and activity surveillance are common functions that many wearable technologies inherently can be tracked by users. Shultz (2017) found that NCAA member institutions that had issued WHOOP bands to athletes were actively monitoring their athletes outside of traditional practice and competition settings. Such surveillance was expressly characterized as more beneficial for athletic departments than athletes themselves:

WHOOP, though, doesn't help the people who actually wear it nearly as much as it does the institutions and coaches who invest in it. Wearing the device puts a student-athlete's personal privacy, standing on the team, and future in pro sports at risk in exchange for what is currently an unverified boost in performance. Continuous-monitoring data can not only indicate that you're not sleeping, but why, and the data are shared with coaches (Shultz, 2017, para. 7).

Given the inadequate, or nonexistent, representation of college athletes in NCAA or athletic department matters, college athletes are often expected and mandated to consent to all dictations of athletic department institutional members (i.e., administrators, coaches, and staff). Such expectation for adherence to expected behaviors is akin to the spontaneous consent mandated within a hegemonic power dynamic (Harry, 2023; Stokowski et al., 2023). Combined with a rigorous schedule and intense social demands, the NCAA as an institutional setting has been characterized as a total institution in which patriarchy is central to continued maintenance and operation (Hatteberg, 2018; Southall & Weiler, 2014). Within this paradigm, college athletes readily consent to athletic department institutional members' dictations believing their best interests are being represented. In the context of the pervasiveness of wearable technology at NCAA member institutions, the stated purpose for adapting technological innovations that track college athletes' biometric data is framed as purely performance related (Gettysburg College, 2020; Holmes, 2020; Lemire, 2022). Such framing lends credibility to the perception that wearable technology is for the benefit of the athletes themselves. However, such technological adaption and the mandated use of wearable technologies are indicative of the modernization of surveillance within the setting of collegiate athletics. Accordingly, the present exploratory study sought to examine the function and pervasiveness of wearable technology implementation in NCAA athletics.

NCAA Hegemony & Paternalism

While the importance of a “father figure” in a child’s life is well accepted by child psychologists (e.g., see Krampe, 2009; Marshall et al., 2001; Meuwissen & Englund, 2016; Rohner & Veneziano, 2001), college sports often embody a metaphorical framework in which the team is a family, the head coach is often the patriarch, and the players—who have almost always reached the age of maturity and are therefore legally adults—are viewed as the coach’s children. Scholars have equated an NCAA athletic department with a paternalistic total institution (Hatteberg, 2018; Southall & Weiler, 2014), expressive of an embedded institutional logic and various structural components, that result in myriad rules and regulations governing the lives of college athletes. Just as a father makes decisions on behalf of his children, coaches, and administrators craft policies that restrict players’ autonomy and economically benefit the organization (Hawkins, 2010). In fact, the NCAA itself places a stated emphasis on protecting athletes from corrupting outside influences (NCAA, 2022). Such benevolent paternalism has long been part of the NCAA’s amateur code and has historically manifested itself in deference to college coaches and administrators among the general public and fans (Byers & Hammer, 1997; Southall & Weiler, 2014).

Hawkins (2010) has characterized the protection of college athletes from “exploitative” commercial practices as a paternal relationship unique to collegiate athletics. In the current collegiate model, coaches assume the role of a father acting in the best interest of their college athlete *children*, who are perceived as unable to make informed autonomous decisions (Corr et al., 2024). If college athletes are indeed children, the adults (e.g., coaches, athletic administrators) are justified in paternalistically exerting

control over many facets of athletes' daily lives (Hatteberg, 2018; Southall & Weiler, 2014). Consequently, to protect their players, college coaches exercise almost total control within their programs. However, this paternalistic control is also the coaches' best defense against loss and humiliation.

Paternalism persists in various forms throughout the institutional field of NCAA athletics. Hawkins (2010) and Southall and Weiler (2014) claim a paternal relationship exists between coaches and college athletes, in general, and Black male profit-athletes¹ specifically. In the current collegiate model, male football and basketball coaches, as agents of the university, athletic department, conference, and NCAA, serve as father figures who act in the best interests of their children (i.e., athletes). Within the collegiate model, traditional grants-in-aid (GIA), as well as more recent forms of compensation, including cost-of-attendance stipends, education-based benefits, and interim name, image, and likeness (NIL) policies, not only reflect adaptation necessitated by legal setbacks to the NCAA's collegiate model but are also designed to protect and preserve a paternalistic code of amateurism.

Within this setting of institutionalized paternalism, and considering the immense financial value of collegiate athletics, the activities and whereabouts of college athletes have historically been monitored through mechanisms such as class checkers (New, 2015; Stokowski et al., 2017). However, the pervasiveness of wearable technology in collegiate athletics offers athletic department institutional members (i.e., administrators, coaches, staff) a viable means to monitor athletes through digital systems (Comeaux, 2018; New, 2015). Within the hegemonic paternalistic power structure of collegiate athletics, such monitorization and surveillance of college athletes exist as a mechanism for control. Given the commercialized nature of the institutional setting of collegiate athletics in the United States (Corr et al., 2022; Southall et al., 2023), an analogous relationship exists between such exerted control and the commodification of college athletes (Corr et al., 2023a).

Athlete Commodification & Surveillance

Within an institutional setting that places an overt emphasis on commercialized logics (Corr et al., 2022; Southall et al., 2023), college athletes serve as distinct *products* with tangible economic value. Institutional stakeholders (e.g., athletic departments, athletic conferences) leverage the value of athletes in pursuit of lucrative corporate partnerships and multimedia rights agreements. Corr et al. (2023a) posited that such commercialized activity effectively commodified competing college athletes according to their value within the institutional setting of collegiate athletics. Given the tangible economic value of college athletes, the hegemonic and paternalistic practices embedded within the framework of collegiate athletics—and the NCAA specifically—are intuitive. In protection of corporate commercial interest, athlete surveillance exists as a normalized practice within various sport spaces and settings (Jones et al., 2016, 2022). As sport and athletics operate as an immensely valuable and consistently growing economic field, organizations interfaced with technology to improve athlete performance and, subsequently, increase the likelihood of competitive success. As competitive success in collegiate athletics is positively

correlated to departmental and institutional revenue generation (Caro, 2012), the consistent emphasis on winning athletic contests embodies the commercialized logic pervasive within the institutional setting of collegiate athletics in the United States.

In the early 2010s, discussions regarding wearable technology use in collegiate athletics started to surface as awareness grew that some teams were potentially gaining a competitive advantage from a device designed by Australian-based company Catapult to monitor physiological and biological indicators of athlete performance. As one strength and conditioning coach referred to it at the time, the Catapult device “is to strength and conditioning what the barbell was 40 or 50 years ago” (Dodd, 2014, para. 12). It is within this setting the perceived impact of wearable technology on increased athlete performance led to institutional-wide adoption of wearable technology devices. Such saturation within the setting of collegiate athletics is normalized, in part, due to the availability of wearable technology in the personal fitness market. Although competitive and recreational athletes alike utilize wearable technologies for performance purposes, extant research has found such *self*-surveillance can have a cumulative negative effect on individuals’ psychological and physiological well-being (Goodyear et al., 2019; McMahon & Penney, 2013).

While athletes engaging in self-surveillance may experience distressful outcomes, athletes surveilled at the organizational level are subject to distinct stressors as well. Jones et al. (2022) posited that surveillance at the direction of sport organizations can result in an expected obedience among surveilled athletes. As the power structure within sport and sport organization explicitly positions head coaches and members of the coaching status at the top of the hierarchy, athletes are likely to consent to surveillance practices normalized throughout the organization. Wearable technology adoption serves as an additional surveillance measure with increasingly invasive biometric tracking capabilities. In the context of sport and athletics, such biometric tracking can be utilized for beneficial and deviant means by coaches that wield a significant level of power over competing athletes (Jones et al., 2016). In the context of the hegemonic paternalistic formulation and operation of collegiate athletics, the adoption and implementation of wearable technology serve to the potential detriment of competing college athletes already subjugated to the will of an organization (i.e., athletic department) and its various members (e.g., coaches, administrators). Correspondingly, the present study sought to examine the use and function of wearable technology within the setting of collegiate athletics in the United States. Given the framework surrounding the power dynamics of collegiate athletics—and sport and athletics, writ large—as well as the normalized surveillance practices embedded within collegiate athletics, the researchers determined current competing college athletes as a primary population to sample.

Methodology

Data Collection

As the study was exploratory in nature, researchers utilized semi-structured interviews with current college athletes to examine the function and pervasiveness of wearable

technology in NCAA athletics. While interviews consisted of current college athletes discussing their experiences with wearable technology, the aim of the study was to explore the pervasiveness of wearable technology in collegiate athletics. Accordingly, the researchers' directed each interview in a manner that focused on a more holistic description of the use of wearable technology among college athletes, teams, and athletic departments. Such a method was deemed appropriate by the researchers given the present study employed a social constructivist epistemological approach. Seeking to determine the manner in which wearable technology is utilized in the setting of collegiate athletics, researchers employed this approach as the utilization and function of wearable technology is determined by and within competing college athletes' lived experiences. The use and function (read: *meaning*) of wearable technology in collegiate athletics is, correspondingly, determined by and influenced due to athlete's unique and distinct experiences.

Interviews ranged from 18 to 30 min in length ($M = 24$ min) and were recorded on a secure audio listening device only accessible to the researchers. Two of the researchers conducted interviews with participants, utilizing the same semi-structured interview protocol in each separate interview. Given the distinctive time demands of competing college athletes, interviews were purposively brief in nature and conducted virtually. In addition, given the present study was exploratory in nature to examine the function and pervasiveness of wearable technology in collegiate athletics, each interviewer asked questions that led participants to describe the use of wearable technology within their respective athletic department, the function of such technology, and the pervasiveness of wearable technology by coaches and teammates.

Participants

The participants in this study consisted of 59 ($n = 59$) athletes currently participating in NCAA athletics as of Fall 2023. Representative primarily of NCAA Division I institutions, participants were identified through a convenience, snowball sampling technique. Initial participants were identified through their direct relationship with the research team (i.e., convenience). Given the researchers in this study have been employed in numerous academic departments at multiple Division I institutions, the initial network of athletes was substantive in nature. Drawing from this sample of personal relationships, researchers utilized a snowball sampling technique to identify other participants. Accordingly, upon the conclusion of an interview with a participant, they were asked if they knew any other current NCAA athletes willing to participate in the study. This word-of-mouth method to secure participants was valuable in establishing a level of trust between researchers and participants. Such sampling technique was valuable in securing additional participants and given the insulated nature of collegiate athletics, is a noted practice in studies examining college athlete populations (Corr et al., 2023b; Kluch, 2023).

The sample of 59 athletes represented ten ($n = 10$) NCAA-sanctioned sports. Thirty-nine ($n = 39$) interview participants currently competed in women's sports and 20 ($n = 20$) competed in men's sports. Table 1 provides anonymized descriptive information of each participant.

Table 1. Characteristics of Participants.

Pseudonym	NCAA classification	Gender	Sport	Issued wearable technology?
Adam	Division I, no football	Men's	Track & Field	No
Allison	Division I, FBS	Women's	Rowing	No
Amy	Division I, no football	Women's	Soccer	Yes
Andre	Division I, FBS	Men's	Football	Yes
Antwan	Division III	Men's	Baseball	No
Ariel	Division I, no football	Women's	Soccer	Yes
Ayesha	Division I, FBS	Women's	Track & Field	No
Beatrice	Division I, FBS	Women's	XC, Track & Field	Yes
Ben	Division I, FBS	Men's	Football	Yes
Carlos	Division I, FBS	Men's	Track & Field	No
Carly	Division I, FBS	Women's	Track & Field	Yes
Caroline	Division I, no football	Women's	Soccer	Yes
Cathy	Division I, no football	Women's	Soccer	Yes
Charles	Division I, no football	Men's	Track & Field	No
Chris	Division I, no football	Men's	Tennis	No
Christina	Division I, no football	Women's	Tennis	Yes
Claire	Division I, FBS	Women's	Soccer	Yes
Clarice	Division I, no football	Women's	Soccer	Yes
Clint	Division I, FBS	Men's	Football	Yes
Cooper	Division III	Men's	Soccer, Tennis	No
Courtney	Division I, FBS	Women's	Tennis	Yes
Darren	Division I, FCS	Men's	Track & Field	No
Deidra	Division I, FBS	Women's	Track & Field	No
Denise	Division I, no football	Women's	Soccer	Yes
Dylan	Division I, FBS	Men's	Football	Yes
Elena	Division I, FBS	Women's	Soccer	Yes
Florence	Division I, no football	Women's	Soccer	Yes
Harriet	Division I, no football	Women's	Tennis	Yes
Jack	Division I, no football	Men's	Track & Field	No
Judith	Division I, FBS	Women's	Tennis	Yes
Julie	Division I, FCS	Women's	XC, Track & Field	Yes
June	Division I, FBS	Women's	Tennis	Yes
Justin	Division I, FBS	Men's	Football	Yes
Kennedy	Division II	Women's	Soccer	Yes
Kimberly	Division I, no football	Women's	Soccer	Yes
Kira	Division I, no football	Women's	Soccer	Yes
Mackenzie	Division I, no football	Women's	Tennis	No
Madison	Division I, no football	Women's	Soccer	Yes
Megan	Division I, no football	Women's	Tennis	Yes
Nicole	Division I, no football	Women's	Tennis	No
Nina	Division I, no football	Women's	Soccer	Yes
Oscar	Division I, no football	Men's	Track & Field	No
Paula	Division II	Women's	XC, Track & Field	No

(continued)

Table 1. (continued)

Pseudonym	NCAA classification	Gender	Sport	Issued wearable technology?
Peter	Division I, FBS	Men's	Football	Yes
Richard	Division I, FCS	Men's	Football	No
Ross	Division I, FBS	Men's	Football	No
Samantha	Division I, no football	Women's	Track & Field	No
Sarah	Division I, FCS	Women's	Tennis	Yes
Sophia	Division I, FBS	Women's	Volleyball	Yes
Sophie	Division I, no football	Women's	Tennis	Yes
Taylor	Division I, no football	Women's	Track & Field	No
Thomas	Division I, FBS	Men's	Track & Field	No
Tia	Division I, FBS	Women's	Soccer	Yes
Timothy	Division I, FBS	Men's	Track & Field	No
Trudy	Division I, no football	Women's	Tennis	No
Ursula	Division II	Women's	Soccer	Yes
Wanda	Division I, no football	Women's	Tennis	Yes
William	Division I, FBS	Men's	Football	Yes
Yolanda	Division I, no football	Women's	Soccer	Yes

Data Analysis

Upon completion of each interview, researchers transcribed audio recordings into textual form and disseminated transcribed documents to participants to verify their responses or expound upon responses they felt needed further clarification. Once all participants' interviews had been transcribed, researchers began the coding process. Given the use of participants' exact language, researchers utilized an in-vivo coding schema to identify pertinent themes across participant interviews. Researchers then performed axial coding to codify emergent in-vivo themes (Creswell et al., 2007), routinely comparing notes to ensure concurrence in prevalent thematic identification (Thompson et al., 1989). Pseudonyms were utilized for each participant and all descriptive information was removed from interview transcripts to ensure anonymity among participants.

At the conclusion of the in-vivo coding process, researchers collaborated to identify primary themes that emerged from the interviews. Given the exploratory nature of this study seeking to examine the function and pervasiveness of wearable technology in collegiate athletics, identified themes lend themselves to these primary points of emphasis; purpose (i.e., function) and usage (i.e., pervasiveness). The conceptualization of these themes was inductive in nature based on the responses and coding of interviews. Accordingly, the researchers conceptualized primary themes in which interview responses provided context to the function and pervasiveness of wearable technology in collegiate athletics: (1) use of wearable technology, (2) effect of wearable technology, and (3) the amount of surveilled activity. The findings section is arranged in correspondence with these primary themes.

Trustworthiness

It is important for researchers conducting and analyzing data gathered through interviews, specifically, to engage in active reflexivity throughout the entirety of the research process (Mauthner & Doucet, 2003). In this sense, the researchers of this study deemed it appropriate to disclose distinct components pertaining to each of their positionality as part of this reflexive process. Such disclosure is an effort for the researchers to engage with their own experiences and identities and ensure transparency within the subsequent findings and discussion sections. Each of the researchers currently serves in a faculty appointment within a sport-related academic discipline and has an extensive professional background in collegiate athletics. While two of the researchers competed in varsity collegiate athletics, two additional researchers are actively involved in athlete development programming at their respective universities. From an epistemological standpoint, the researchers lived experiences participating and interacting with current college athletes led to the formulation of this exploratory study. As indicated previously, initial interviews were conducted with participants individually recruited by the research team based on previous relationships. To ensure consistency, each of the researchers participated in the various rounds of coding, routinely comparing notes in an effort to establish consensus regarding in-vivo themes. While each of the researchers possess a comprehensive background working in collegiate athletics and with college athletes, the researchers took appropriate steps to ensure consistency within the reported findings below.

Findings

Of the participants in this study, 37 (63%) indicated they had been issued wearable technology while competing as a collegiate athlete. Issued technology varied from standard heart rate monitors, global positioning system (GPS) devices, and smart rings/bracelets (e.g., *Oura*, *WHOOP*). The wearable technology product utilized by most participants was *Catapult* vests. Of the 18 participants that had been issued *Catapult* vests, 12 competed in men's or women's soccer and 6 competed in football. Although 22 participants indicated they had not been issued any wearable technologies as a collegiate athlete, nearly 80% of participants ($n = 46$) indicated they felt monitored by their coach and/or athletic department.

Use of Wearable Technology

Participants indicated that wearable technology was often utilized for performance-related purposes. Various participants, spanning multiple sports, detailed the use of biometric data collected from wearable technologies to track performance metrics (e.g., distance run, calories burned, heart rate) during practice and competition. Another commonly emphasized component of wearable technology usage was coaches and trainers' use of biometric data to prescribe recovery routines post-workouts and -games. Cathy, a Division I women's soccer player, summated the use of biometric data to assist coaches in developing training plans based off players' physiological expenditures during practice sessions:

If we have a high intensity session one day and our loads are high, then they can alter the next session. Or if our loads are getting high during a session, then they can end [practice] early.

Sarah, a Division I women's tennis player, stated similarly that a primary use of wearable technology was "to monitor training load and to prevent overtraining." Such performance-related functions are indicative of the value that wearable technology can provide to both coaches and athletes within a sport setting but also, and perhaps most saliently, reveal the nature of wearable technology as it is intended to be utilized; to improve athletic performance. In the context of sport and athletics, where improved athlete performance and competitive success are manically pursued, such utilization of wearable technology for competitive purposes is unsurprising.

While performance tracking was a stated use and benefit of wearable technology, many participants detailed coaches' use of wearable technologies to ensure players were adequately exerting themselves during training sessions and competitions. The term "effort level" was used by numerous participants to codify the use of wearable technologies to ensure players were training at a level accepted by coaches and staff (e.g., strength & conditioning staff). Beatrice, a Division I women's cross country and track & field runner, stated that coaches would "adjust our workouts depending on effort" as measured by heart rate monitors issued to runners on the team. Elena, a Division I women's soccer player, expressed similarly that coaches would use heart rate monitors and GPS devices to track "if we push ourselves enough in practice." In this sense, the use of wearable technology serves as a mechanism in which to track performance-related metrics and monitor the physical exertion levels of athletes themselves. Accordingly, wearable technology provides coaches the ability to make a determination of the level of physical stress an athlete endures during training or competition irrespective of an athlete's sentiment. Such findings are consistent with extant literature (Jones et al., 2016) and embody the potential deviant use of wearable technology in sport organizations. Under the guise of improving athlete performance, coaches are able to leverage their distinct organizational power to test the physical exertion limits of competing athletes when tracking biometric statistics. Such actions dehumanize an athletes' performance and promulgate the significance of a number (e.g., heart rate, distance traveled) measured via biometric tracking (Goodyear et al., 2019; McMahon & Penney, 2013).

Effect of Performance Tracking on Team Dynamics

Twenty-three participants who were issued wearable technology indicated that their individual biometric data was communicated and shared with them in a group (i.e., team) setting. Ben, a Division I football player, stated that biometric data was posted in team areas (e.g., weight room, locker room) "to incite competition. Data like top speed being posted [sic] in the weight room for everybody to see." While the majority of participants were unaware of performance technology's effect on team dynamics, three participants indicated that coaches sharing biometric data in a

group setting did foster dissent among teammates. Tia, a Division I women's soccer player, discussed a scenario in which coaches placed a video monitor in the center of the locker room and regularly uploaded every player's biometric data after workouts, practice, and games. She indicated that the coaching staff used this data as a metric to display which athletes were the most committed to both their teammates and the team's success that season. Tia explained that divisions formed in the locker room between players that were "ranked" higher based on their physical performance output (e.g., distance run, calories burned). She also indicated that coaches would routinely shame players who were ranked lower in these physical performance categories and issue punishments for these athletes (e.g., running after practice, limiting playing time).

Other participants also indicated that wearable technology devices sowed division in the locker room. Playing at a Group-of-5 football program, Ben indicated that limited financial resources meant the team only had 25 Catapult vests. These 25 vests were worn almost exclusively by the starting offensive and defensive players. Accordingly, Ben stated that being issued a Catapult vest by the training staff became a status symbol representing your importance to the team as a starter. While coaches may often seek to create competition among teammates, the use of wearable technology and/or the presentation of biometric data may serve to foster undue discord between teammates.

Surveilled Activity

Risks of Athletic Performance Surveillance. From a performance and athletic training standpoint, the use of wearable technology to monitor practices and competitions was nearly ubiquitous. Seemingly, athletic programs are implementing wearable technologies to track athletes' physical performance and exertion in both on- and off-season settings. Responses concerning monitoring athletic performance were relatively consistent among participants, regardless of sport played. Athletes indicated that coaches implemented wearable technologies during training and competition to monitor a number of medical and performance-related metrics. Overall, athletes viewed the use of wearable technology *explicitly* for athletic performance as valuable within their training schedule and physical development. However, a number of participants indicated that the implementation of biometric data tracking posed significant risks to college athletes' wellbeing, risks largely dependent on the tracking program implemented by members of the coaching staff. Elena, a Division I women's soccer player, crystalized this point:

I think that technology is a great way to help [college athletes] monitor our performance and help us improve it, especially for recovery. However, coaches might not always use it the right way. I think it depends on the school and the real purpose of why coaches want their athletes to use [wearable technology].

Courtney, a Division I women's tennis player, also discussed the duality between wearable technology's benefits and potential detriments:

I think [wearable technology] can be helpful and useful to see recovery, but our coaches don't even work us less hard when they do see that we had really poor sleep the night before, so it kind of counteracts the point.

Regarding the potential risks of wearable technology and biometric data tracking, June, a Division I women's tennis player, stated much more plainly, "Wearable technologies, especially ones that track location and sleep patterns, are invasive and psychologically damaging to college athletes."

Location & Sleep Surveillance. While the use of wearable technology for biometric data tracking during athletic training or competition is noteworthy, nearly 80% of participants ($n = 46$) indicated they felt routinely monitored by athletic department institutional members (i.e., coaches, trainers, staff) both in and out of athletics specific settings. For many participants, wearable technology issued by coaches or members of the training staff was directly utilized to monitor their whereabouts. Beatrice, a Division I women's cross country and track & field runner, noted that athletes on her team felt constantly scrutinized stating, "the staff control us somehow." Jack, a Division I men's track & field athlete, noted the scrutiny college athletes face from their coaches as well saying, "with social media connections and accounts, there is almost no privacy." Timothy, a Division I men's track & field athlete, detailed the physical tracking of athletes, "my coach would take body weight either daily or weekly, depending on the athlete, and would indirectly monitor our diets." While not forms of biometric tracking through wearable technology, athletes participating in this study identified that they felt regularly monitored by coaches and athletics staff throughout their day-to-day activities. Oscar, a Division I track & field athlete, exemplified the feeling of being monitored constantly by coaches, "as an athlete, coaches and staff monitor every public step in [your] life. From your grades and performance, through practice, to how you walk around and interact with teammates and friends on campus." June, a Division I women's tennis player, also added:

Everything you eat. How many extra workouts, practices you do. How you sleep...is always monitored at some level, even if that is just in the form of a question: 'how did you sleep last night?'

Such statements lend themselves to the significant investment athletic departments make in college athletes as well as the social status college athletes embody on college campuses across the country. Within the context of such investment, wearable technology is an increasingly utilized and valuable mechanism in which coaches, staff, and athletic administrators are able to effectively and efficiently *protect* their investments (i.e., college athletes).

The use of wearable technology to track athletes outside of athletic specific settings (e.g., practice, competition) was pervasive throughout participant responses. Clint, a Division I football player, stated that thanks to wearable technology issued by the athletic department, "every move you make is under surveillance, on and off the field."

Justin, another Division I football player, stated that wearable technology was issued to athletes to track “where you were at all times away from the [football] facility.” Mackenzie, a Division I women’s tennis player, added, “coaches are able to see everything” while June, another Division I women’s tennis player, described the technological interface utilized to geolocate athletes’ whereabouts:

I know that they’ve used [technology] for class tracking. They have monitors in select number of classrooms on campus and when [an athlete’s] phone reaches its tracking distance, it records that they’re in class...a lot of [athletic departments] use something like the Polar watch or WHOOP and say that since it’s a university owned item that they need to see everything.

In addition to ensuring class attendance, June noted that athletes were routinely tracked in off-campus environments as well, stating, “[coaches track] sleep and even call [athletes] out the next day if they saw that they drank the night before or went out to the bars.”

While, overall, physical location tracking was infrequent among participants, sleep tracking was stated to be a common metric monitored by coaching staffs. Julie, a Division I cross country and track & field runner, stated, “my coach tracks my runs as well as my sleep.” Among the participants that had been issued wearable technology to track their sleep and rest, many believed it was for their direct benefit. Sarah, a Division I women’s tennis player, stated that biometric tracking was “to help me to be a better athlete...[to] make sure that my body is at the best level of athleticism.” Denise, a Division I women’s soccer player, stated practices such as location and sleep monitoring were accepted “because [coaches] just monitor specific things for our benefit, not in a stalking or strict way.” Such surveillance is Orwellian in nature as athletic departments, by way of organizational members (e.g., coaches, administrators), are able to actively monitor various components of competing athletes lives through biometric tracking. Given the prioritization of athletic performance, such monitoring of athletes’ biometric statistics provides athletic departments with increased control over athletes during important recovery periods. Wearable technology allows members of the coaching staff invasive control over sleep and recovery, physiological vital aspects impacting athlete performance.

In the context of hegemonic structures, in which spontaneous consent is a centralized component to the hegemon’s consolidation of power, the number of participants indicating location and sleep track were for the expressed benefit of athletes is noteworthy. In fact, many participants indicated they would consent to additional degrees of biometric tracking then they were already exposed too. Timothy, a Division I track & field athlete, stated, “wearable tech is a great way to effectively monitor athletes and it should be used more...if [an] athlete is known to cause trouble then I believe [tracking] should be a last resort option.” Oscar, another Division I track & field athlete, stated “the tracking of athlete whereabouts was a good idea” while Carlos, a Division I track & field athlete as well, stated “I really like the idea of [biometric tracking] and hope that it gets used more often in the future.” Each of these responses is

indicative of the readiness at which college athletes participating in this study willingly consent to monitoring programs. Such willful consent may be the manifestation of the paternalistic relationship between coaches and athletes prevalent in many collegiate athletics programs. Ben, a Division I football player, summated the trust athletes bestow upon coaches in the context of college athletes' willingness to consent to the practice of biometric tracking from wearable technologies; "you grow up with the assumption [coaches] know what's best. They're using science to make you better." College athletes' consent to biometric tracking for non-performance-related reasons, seemingly, serves as a direct reflection of the patriarchal paradigm in college sport and is consistent with extant scholarship positing athletes are likely to consent to increased surveillance practices as they are continuously normalized (Jones et al., 2022).

Despite being provided wearable technology by her athletic department, Christina, a Division I women's tennis player, stated financial inhibitors limited the type of product she was provided:

I wanted my team to use WHOOP or something similar, but our athletic department didn't allow my coach to use his budget for those. I thought it would be good to track calories, sleep, recovery, et cetera but I also feared that it could be used against me at any time. It's a weird middle ground. If I can use [wearable technology] for me, then I'm good. If it's used against me, it's a no go.

In instances where financial limitations prevented coaches from issuing wearable technology to track athletes' outside of practice or competition settings, participants indicated coaching staffs used other mechanisms to regularly and routinely check in on their athletes. Cathy, a Division I women's soccer player, described a "readiness form" that each member of the team was required to submit to the coaching staff every morning during the season; "we complete questions based on how we are feeling physical and mentally." While location tracking allows coaches to geolocate athletes to ensure they are attending class or at home, some participants indicated that coaches or staff physically appeared in classrooms or tutoring sessions to verify athlete attendance. Such monitoring practices have been commonplace in collegiate athletics (Bradley, 2017; Holmes, 2020; Shultz, 2017) and contribute to the theoretical perception of college sport as a total institutional setting. However, for athletes that were being tracked by GPS through institutionally issued wearable technology, the adaption of technology within collegiate athletic programs is indicative of the modernization of college athlete monitoring.

Discussion & Implications

Programmatic implementation of mandated wearable technology for college athletes for both revenue generation and performance-related measures can manifest in distressful outcomes (Goodyear et al., 2019; Jones et al., 2016; McMahon & Penney, 2013). The singular focus on revenue acquisition in Division I athletes aligns with extant

literature examining a hyper-commercialized logic permeated throughout the NCAA, and perpetuated by the NCAA itself (Corr et al., 2022; Southall et al., 2023). However, coaches and staff use of athletes' biometric data can have direct outcomes on intra-team dynamics as well. The use of biometric data as a comparison point between teammates creates a unique hierarchical environment that categorizes athletes based on biometric measures (e.g., target heart rate, distance traveled, calories burned; Jones et al., 2016). While the use of wearable technologies to track such biometric variables is of distinct value (Bourdon et al., 2017; Cardinale & Varley, 2017; Luczak et al., 2020), the tendency for college coaches to disseminate such personal performance data within a team setting may be detrimental to team dynamics and unity. In practice, the presentation of biometric and tracking data in group settings also raises significant questions regarding college athletes' right to personal privacy (Jones et al., 2016, 2022; Shultz, 2017).

While most Division I athletic departments feature budgets that can accommodate the price of innovation with regard to wearable technology, athletic programs operating within budgetary constraints have developed cost efficient monitoring programs to measure athletes' performance data. The implementation of a mandated daily self-reporting system as indicated by respondents in this study is a byproduct of an athletic department's inability to afford wearable technologies designed for tracking (e.g., WHOOP) but a coaching staff's desire to still monitor athletes. As such, financial constraints do not seemingly limit a coach's desire or ability to implement a program to monitor the biometric characteristics of athletes. Similar to traditional class checkers or monitors (New, 2015; Stokowski et al., 2017), the desire among athletic department institutional members to surveil athletes is only exacerbated and made more efficient with the distribution of wearable technology. Within the context of a hegemonic and paternalistic setting, college athletes readily consent to this surveillance out of a belief that coaches and staff hold their best interests. Accordingly, consistent with the supposition of Jones et al. (2022), as wearable technology implementation becomes increasingly normalized over time, college athletes are more likely to comply with organizational demands. In addition, the patriarchal construction of the coach-athlete relationship is directly communicated through comments of participants in this study like that of Ben, a Division I FBS football player: "you grow up with the assumption [coaches] know what's best." This assumption manifests in the passively consensual acceptance to be surveilled through the use of wearable technology as a college athlete and is overtly indicative of the distinct power dynamic between athletes and coaches.

Within the hyper-commercialized setting of NCAA athletics, and although college athletes are permitted to monetize their NIL, college athletes forgo control of their personal biometric data. Much of this relinquished autonomy can be attributed to patriarchal hegemony within the total institutional setting college athletes exist. Accordingly, the absolute authoritative position held by college coaches can be yielded to enforce biometric data tracking and engage in technological surveillance. Such practices have been commonplace across the spectrum of NCAA Divisions and will likely be utilized by coaches to a greater extent as wearable technologies become less cost prohibitive and societally accepted as integral to establishing an informed, and thereby

credible, training regimen. With inadequate representation and little to no room for dissent, college athletes will likely continue to spontaneously consent to the collection and repurposing of their personal biometric performance data.

As college sports becomes further entangled with sport-wagering companies, the NCAA should consider taking steps to mitigate many athletic departments' insatiable desire to monetize each product at their disposal. Such hyper-commercialization is inclusive of college athletes themselves as biometric data collected from university-issued wearable technology is permitted to be sold to sport wagering companies to assist in establishing more informed betting lines, a practice similar to mandating athletic programs produce injury reports on a timely basis (Dodd, 2022; Perez, 2022). These reports (i.e., injury reports) are further indicative of the pervasiveness of gambling within the college sport marketplace as availability reports clarifying the physical health status of Big Ten athletes will assist sportsbooks in establishing more informed betting lines (Vannini, 2023). As the demarcation between college and professional sport becomes more and more vague, college athletes possessing inadequate or non-existent representation are mandated to adhere to the rules, regulations, and practices set forth by varying institutional- (e.g., conference administrators, NCAA officials) and organizational members (e.g., athletic administrators, coaches).

Conclusion & Future Research

Multiple participants commented on the use of wearable technology for location-tracking purposes. Respondents also commented on the use of technology in general by athletic department institutional members to track and monitor their location remotely. A commonly implemented program mentioned by participants was the application *Spotter*. Geo-location applications such as *Spotter* allow the tracking of technological devices (i.e., cell phones from a remote location). While not a form of wearable technology, per se, the mandatory use and monitorization of athletes' geo-location through the implementation and use of an application such as *Spotter* is indicative of the use of technological innovation for the purposes of surveillance. Interestingly, considering the paradigm of collegiate athletics within the setting of higher education, the *Spotter* application is described by the developer as "User-friendly GPS trackers and GPS watches for children, the elderly, animals and valuables" (Spotter, 2023).

The *Spotter* application developer also states that users' locations and personal data will not be shared with third parties unless consent is given by the user themselves or stances involving a minor (i.e., parent tracking child's device). The researchers decided to reach out via public records request to NCAA member institutions of athletes that participated in the study. Of the eight institutions that responded to the public records request, none stated any consensual agreement between athletes and the university, athletic department, or coaches existed to allow athletic department, institutional members to track the location of athletes' personal devices. Future legal examination of location tracking, and the non-consensual relationship between athletic departments and athletes themselves, is necessary to codify the legality of the relationship regarding biometric and geo-location tracking between these two parties.

As technology is constantly evolving, the manner in which pervasive technological devices are utilized and implemented remains a valuable area of inquiry. Future scholarship examining the impact of wearable technology for the purposes of monitorization and surveillance is of both practical and theoretical importance. While college athletes are subjected to, and directly a part of, an institutional setting that limits their autonomous decision making (Hatteberg, 2018; Southall & Weiler, 2014), the findings of this study illustrate the capacity in which wearable technology is currently utilized as a tool of control over college athletes. Such findings reinforce the framework of the NCAA as a total institution and also lend themselves to the paternalistic control asserted by coaches and athletic administrators.

Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The authors received no financial support for the research, authorship, and/or publication of this article.

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Note

1. Profit-athletes are college athletes whose estimated fair market value exceeds the value of NCAA-approved direct university athletic department compensation (e.g., grant-in-aid, cost-of-attendance stipend, education-related “Alston” award).

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